

HARIOM KHONDE

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EDUCATION

PES University RR Campus | 7.92

Bachelor of Technology in Computer Science and Engineering

Aug 2023 - Jul 2027

Bengaluru, India

Shahu Maharaja pu science college | 93.33%

Pre-University Course (PCM-100%)

2023

Bidar, India

EXPERIENCE

CHEAL Research Center

Research Intern

Jun 2025 - Jul 2025

Pes University

- Designed real-time body-mounted monitoring system for medication adherence in geriatric diabetic patients
- Integrated **YOLOv8** object detection and **OCR** (PaddleOCR) for medicine packet recognition achieving 92% accuracy
- Built complete AI pipeline using **Python**, **OpenCV**, **ultralytics** with custom dataset training and fallback recognition

TECHNICAL SKILLS

Languages: C++, Java, C, JavaScript, Python

Technologies/Frameworks: React.js, Express.js, Node.js, MySQL, Redis, Mongoose

Tools: Git, GitHub, Docker, Hadoop, Kubernetes, AWS(basics), MongoDB

PROJECTS

YAK - Yet Another Kafka | Github | Python, FastAPI, Redis, Distributed Systems

2025

- Engineered fault-tolerant distributed message broker with **zero data loss guarantee** using Redis-based atomic leader election and synchronous replication across broker nodes
- Implemented High Water Mark (HWM) consensus protocol ensuring messages committed only after follower acknowledgment, preventing inconsistent reads during leader failures
- Built intelligent producer-consumer clients with automatic leader discovery and seamless failover, successfully demonstrating complete system recovery after catastrophic leader crashes

Brain Tumor MRI Analysis System | Github | React, Python, TensorFlow, Flask

2024

- Developed AI-powered web application using **EfficientNetB4** to classify brain tumors with **91.1% accuracy** across 3,000+ MRI images
- Built full-stack solution with React TypeScript frontend and Flask backend serving TensorFlow/Keras models
- Integrated **Google Gemini AI** for treatment recommendations and **Grad-CAM visualizations** achieving F1-score of 0.968

SmartWallet — Wallet Management System | Github | Node.js, Express, MySQL

2025

- Architected a full-stack digital wallet with **atomic P2P and merchant payments** via MySQL stored procedures, preventing partial-update failures and duplicate charges through transaction hash idempotency
- Designed **5+ triggers and stored procedures** with wallet audit trail, bcrypt PIN hashing, JWT auth, and role-based access for users and merchants
- Built automated reward expiry via **MySQL scheduled events** and a monthly analytics engine with CSV export, chart visualizations, and hourly refund retry logic

HACKATHONS & COMPETITIONS

- CODE RED'25 Hackathon** - Developed a waste management platform with AI categorization
- Epoch 2.0 DataThon** - Built the Tumor-Detect AI solution for brain tumor MRI analysis
- Terrathon 4.0** - Created the EcoScan mobile app promoting sustainable consumption using AI
- Actively participating in various hackathons and coding challenges to enhance technical skills